



Prayas JEE 2.0 (2024)

Quadratic Equations

DPP-08

- Let $P(x) = x^2 + bx + c$, where b and c are integer. If $P(x)$ is a factor of both $x^4 + 6x^2 + 25$ and $3x^4 + 4x^2 + 28x + 5$, find the value of $P(1)$.
- If $\forall p \in R$ one root of the equation $x^2 + 2px + q^2 - p^2 - 6 = 0$ is less than 1 and other root is greater than 1, then range of q is-
 (1) $(-\infty, -2)$ (2) $(-2, 2)$
 (3) $(-\sqrt{5}, \sqrt{5})$ (4) $(2, \infty)$
- Let r_1, r_2, r_3 , be roots of equation $x^3 - 2x^2 + 4x + 5074 = 0$, then the value of $(r_1 + 2)(r_2 + 2)(r_3 + 2)$ is
 (1) 5050 (2) -5050
 (3) -5066 (4) -5068
- Let a, b, c are roots of equation $x^3 + 8x + 1 = 0$, then the value of

$$\frac{bc}{(8b+1)(8c+1)} + \frac{ac}{(8a+1)(8c+1)} + \frac{ab}{(8a+1)(8b+1)}$$
 is equal to
 (1) 0 (2) -8
 (3) -16 (4) 16
- If $\alpha, \beta, \gamma, \delta$ are the roots of equation $x^4 - \beta x + 3 = 0$, then the equation whose solutions are $\frac{\alpha + \beta + \gamma}{\delta^2}, \frac{\alpha + \beta + \delta}{\gamma^2}, \frac{\alpha + \gamma + \delta}{\beta^2}, \frac{\beta + \gamma + \delta}{\alpha^2}$ is-
 (1) $3x^4 - \beta x^3 - 1 = 0$
 (2) $3x^4 - \beta x^3 + 1 = 0$
 (3) $3x^4 + \beta x^3 - 1 = 0$
 (4) $3x^4 + \beta x^3 + 1 = 0$
- Let α, β and γ are the roots of the cubic $x^3 - 3x^2 + 1 = 0$. Find a cubic whose roots are $\frac{\alpha}{\alpha-2}, \frac{\beta}{\beta-2}$ and $\frac{\gamma}{\gamma-2}$. Hence or otherwise find the value of $(\alpha-2)(\beta-2)(\gamma-2)$.
- If the sum of two of the roots of $x^3 + px^2 + qx + r = 0$ is zero, then $pq =$
 (1) $-r$ (2) r
 (3) $2r$ (4) $-2r$
- If α, β and γ are the roots of equation $x^3 - 3x^2 + x + 5 = 0$ then $y = \sum \alpha^2 + \alpha\beta\gamma$ satisfies the equation
 (1) $y^3 + y + 2 = 0$
 (2) $y^3 - y^2 - y - 2 = 0$
 (3) $y^3 + 3y^2 - y - 3 = 0$
 (4) $y^3 + 4y^2 + 5y + 20 = 0$
- If $P(x)$ is divided by $x - 7$ gives remainder 5, and when divided by $x - 5$ gives remainder 7. What will be the remainder when $P(x)$ is divided by $(x^2 - 12x + 35)$?
 (1) 35 (2) -35
 (3) $x - 35$ (4) $12 - x$
- If α, β & γ are the roots of the equation $x^3 - x - 1 = 0$ then, $\frac{1+\alpha}{1-\alpha} + \frac{1+\beta}{1-\beta} + \frac{1+\gamma}{1-\gamma}$ has the value equal to:
 (1) zero (2) -1
 (3) -7 (4) 1

[Challenger]



Note: Kindly find the Video Solution of DPPs Questions in the DPPs Section.

Answer Key

1. (4)
2. (2)
3. (2)
4. (3)
5. (4)

6. (3)
7. (2)
8. (2)
9. (4)
10. (3)



PW Web/App - <https://smart.link/7wwosivoicgd4>

Library- <https://smart.link/sdfez8ejd80if>